Nine Ways to Reduce Cognitive Load in Multimedia Learning

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What is Multimedia Learning and Instruction?

- **Multimedia learning***:* learning from words and pictures.
- **Multimedia instruction***:* presenting words (printed or spoken) and pictures (static or dynamic) that are intended to foster learning.
- An important example of multimedia instruction is a **computer-based narrated animation**.

What is Meaningful Learning?

- **Meaningful learning***:* deep understanding of the material, which includes:

✓ **attending to** important aspects of the presented material

✓ mentally **organizing** it into a **coherent cognitive structure**, and

✓ **integrating** it with relevant existing knowledge.

Meaningful learning

Meaningful learning

- is reflected in the ability to apply what was taught to new situations
- involves the construction of a mental model.

How to measure learning outcomes?

- retention test
- by using problem-solving transfer tests.

Cognitive Load & Cognitive Overload

- **Cognitive Load (CL)**: Meaningful learning requires **substantial cognitive processing** during learning, but the learner's **capacity** for cognitive processing is **limited**.
- **Cognitive overload**: the learner's **intended** cognitive processing **exceeds** the learner's **available** cognitive capacity.

How the Mind Works

- **Three assumptions** based on research in cognitive science:
- **First, the human information-processing system consists of two separate channels —**

✓ an **auditory/verbal channel** for processing auditory input,

✓ a **visual/pictorial channel** for processing visual input and pictorial representations.

(Paivio 1986, Baddeley 1998, Mayer 2001)

How the Mind Works

- **Second, each channel in the human information-processing system has limited capacity.**

  (Chandler and Sweller 1991, Sweller 1999, Baddeley 1998)

- **Third, meaningful learning requires a substantial amount of cognitive processing to take place in both channels.**

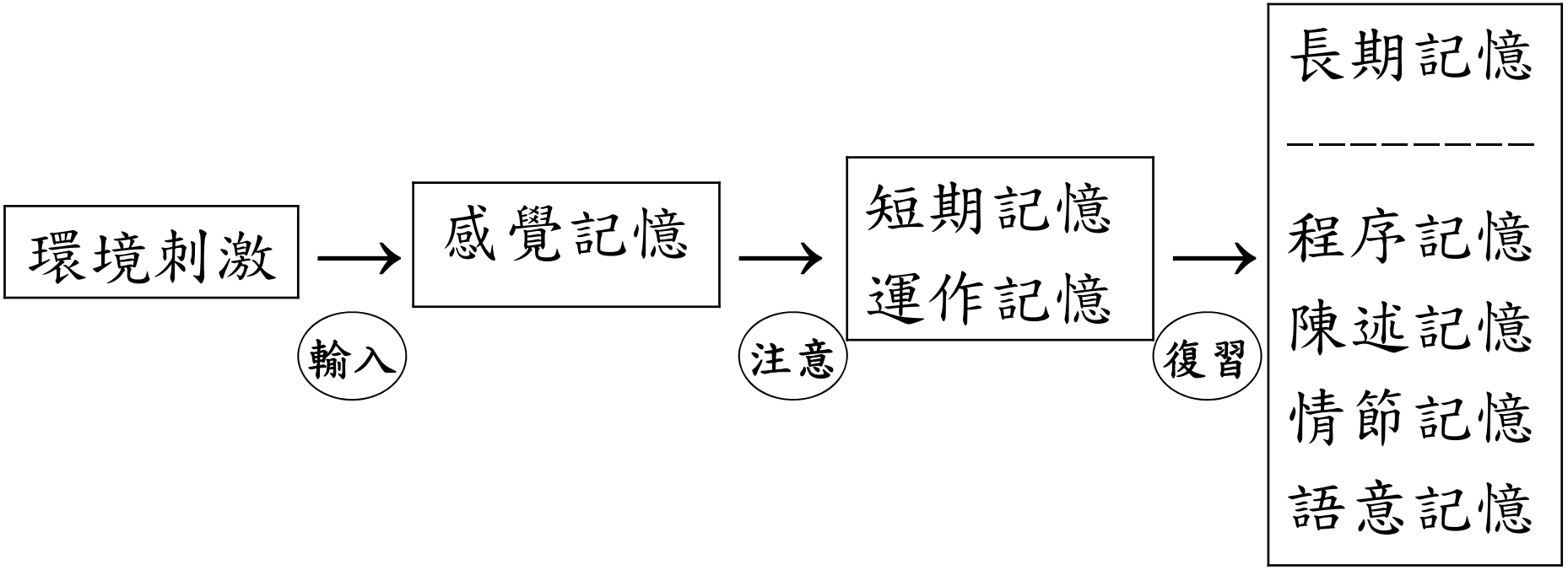
  (Wittrock 1989, Mayer's 1999, 2002)

How the Mind Works

TABLE 1: Three Assumptions About How the Mind Works in Multimedia Learning

Assumption	*Definition*
Dual channel	Humans possess separate information processing channels for verbal and visual material
Limited capacity	There is only a limited amount of processing capacity available in the verbal and visual channels.
Active processing	Learning requires substantial cognitive processing in the verbal and visual channels.

訊息處理模式



How the Mind Works

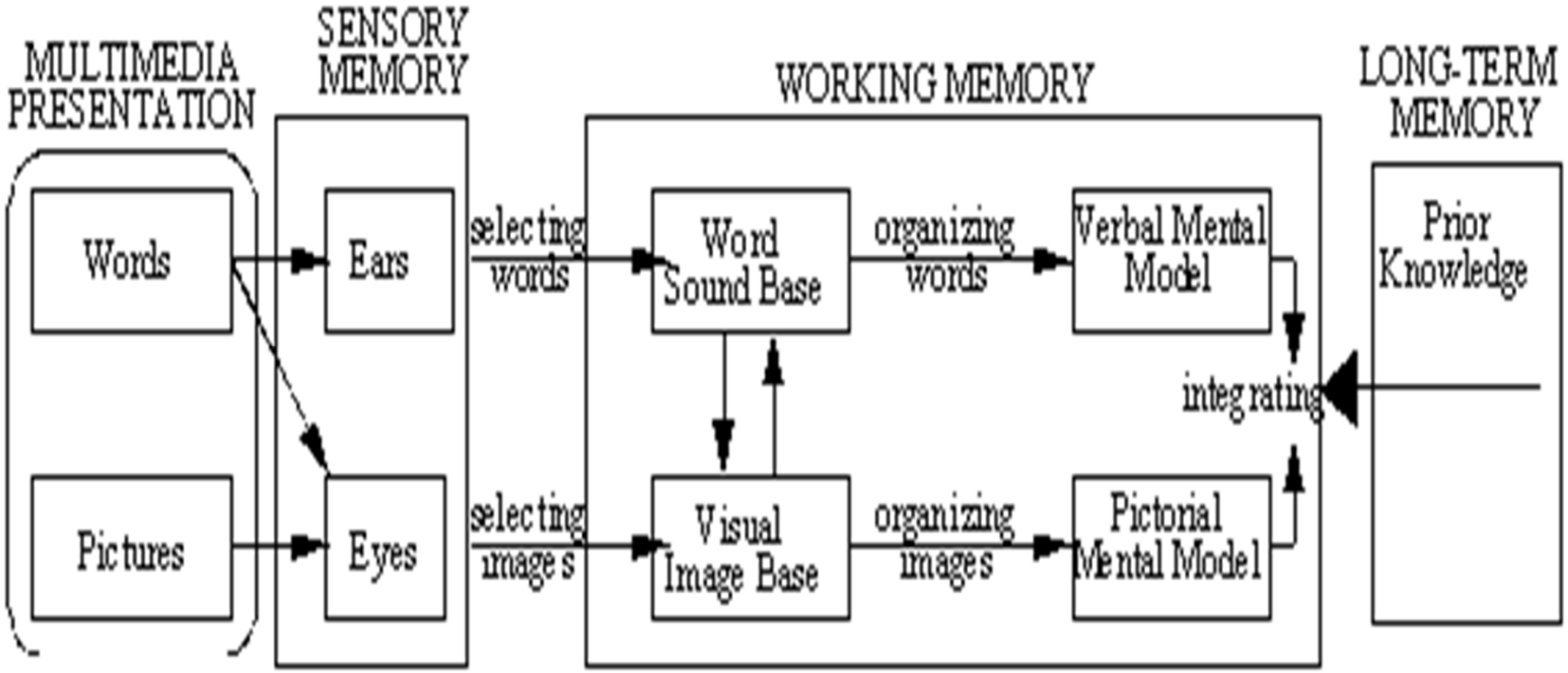


FIGURE 1 Cognitive theory of multimedia learning.

The Case of Cognitive Overload

Three Kinds of Cognitive Demands

- The potential for **cognitive overload** is a **central challenge** for instructors and learners.

- We distinguish among three kinds of cognitive demands (三種認知需求):

✓ **Essential processing** (必要的處理)

✓ **Incidental processing** (附帶的、非主要的處理)

✓ **Representational holding** (表徵保留)

The Case of Cognitive Overload

Three Kinds of Cognitive Demands

- **Essential processing** cognitive processes that are required for making sense of the presented material, such as the five core processes mentioned. （理解所呈現的材料）
- **Incidental processing** cognitive processes that are not required for making sense of the presented material. (ie, background music). （無需理解的材料，如背景音樂）
- **Representational holding** cognitive processes aimed at holding a mental representation in working memory over a period of time. （在工作記憶區保留的心理表徵）

The Case of Cognitive Overload

Three Kinds of Cognitive Demands

TABLE 2
Three Kinds of Demands for Cognitive Processing in Multimedia Learning

Type of Processing	*Definition*
Essential processing	Aimed at making sense of the presented material including selecting, organizing, and integrating words and selecting, organizing, and integrating images.
Incidental processing	Aimed at nonessential aspects of the presented material.
Representational holding	Aimed at holding verbal or visual representations in working memory.

Total Processing Intended = Essential processing + Incidental processing + Representational holding.

The Case of Cognitive Overload
When does cognitive overload occur?

- Cognitive overload occurs when the total intended processing exceeds the learner's cognitive capacity. (認知負荷發生在需處理的材料超過認知容量)

The Case of Cognitive Overload
How to reduce cognitive load?

- Reducing cognitive load can involve redistributing essential processing, reducing incidental processing, or reducing representational holding.

Five Scenarios of Cognitive Overload

- Type 1 - Essential processing in visual channel > cognitive capacity of visual channel
  (solution: 1.off-loading)
- Type 2 - Essential processing (in both channels) > cognitive capacity
  (solution: 2.Segmentation, 3.Pre-training)
- Type 3 - Essential processing + incidental processing (caused by extraneous material) > cognitive capacity
  (solution: 4.Weeding, 5.Signaling)
- Type 4 - Essential processing + incidental processing (caused by confusing presentation) > cognitive capacity
  (solution: 6.Aligning, 7.Eliminating redundancy)
- Type 5 - Essential processing + representational holding > cognitive capacity (solution: 8.Synchronizing, 9.Individualizing)

Type 1 - Essential processing in visual channel > cognitive capacity of visual channel

- Problem: **One channel is overloaded** with essential processing demands.

- Solution 1: **Off-loading**: Move some essential processing from visual channel to auditory channel.
- One solution to this problem is to present words as narration.

Type 1 - Essential processing in visual channel > cognitive capacity of visual channel

- Test the effectiveness of **Off-loading:**
- In a series of six studies at UCSB, students **performed better** on tests of problem-solving transfer when scientific explanations were presented as **animation and narration** rather than as **animation and on-screen text** (Mayer & Moreno, 1998 & 1999; Moreno, Mayer, Spires, &Lester, 2001).（動畫與文字解說組及動畫與螢幕文字組，前者的問題解決表現較佳）
- The median effect size was 1.17.
- This result is referred to as a **Modality Effect.**

Type 2 - Essential processing (in both channels) > cognitive capacity

- Problem: **Both channels are overloaded** with essential processing demands in Working Memory.
- Solution 2: **Segmenting:** Allow time between successive bite-size segments.
- Solution 3: **Pre-training:** Provide pre-training in names and characteristics of components.

Type 2 - Essential processing (in both channels) > cognitive capacity

- Test the effectiveness of **Segmenting**:
- Researchers broke a narrated animation explaining lightning formation into 16 segments. After each segment was presented, the learner could start the next segment by clicking on a button labeled *CONTINUE*. **Students who received the segmented presentation performed better** on subsequent tests of problem-solving transfer **than** did students who received a **continuous presentation.** （分段動畫呈現方式與連續動畫呈現方式，前者的問題解決表現較佳）
- The median effect size was 1.36.
- This result is referred to as a **Segmentation Effect**.

Type 2 - Essential processing (in both channels) > cognitive capacity

- Test the effectiveness of **Pre-training**:
- In a series of three studies involving narrated animations about how brakes work and how pumps work, **students performed better** on problem-solving transfer tests **when the narrated animation was preceded by a short pre-training** about the names and behavior of the components（在課前預先簡短教學關於要件的名字與功能，學生的問題解決表現較佳）
- The median effect size was 1.00.
- This result is referred to as a **Pre-training Effect**.

Type 3 - Essential processing + incidental processing (caused by extraneous material) > cognitive capacity

- Problem: **One or both channels are overloaded** by essential and incidental processing demands (caused by extraneous material).
- Solution 4: **Weeding:** Eliminate interesting but extraneous material to reduce processing of extraneous material.
- Solution 5: **Signaling:** Provide cues for how to process the material to reduce processing of extraneous material.

Type 3 - Essential processing + incidental processing (caused by extraneous material) > cognitive capacity

- Test the effectiveness of **Weeding**:
- In a series of five studies carried out in our laboratory at UCSB, students performed better on problem-solving transfer tests after receiving a concise narrated animation than an embellished narrated animation (Mayer, Heiser, & Lonn, 2001; Moreno & Mayer, 2000).（使用精簡後的敘述動畫，學生的問題解決表現較佳）
- The median effect size was 0.9.
- This result is referred to as a **Coherence Effect**.

Type 3 - Essential processing + incidental processing (caused by extraneous material) > cognitive capacity

- Test the effectiveness of **Signaling**:
- Mautone and Mayer (2001) constructed a 4-min narrated animation explaining how airplanes achieve lift. In this study they conducted on signaling and found that students who received the signaled version of the narrated animation performed better. (在解釋飛機起降的動畫中，提供符號指示可使得學生表現較佳)
- The effect size was 0.74.
- This result is referred to as a **Signaling Effect**.

Type 4 - Essential processing + incidental processing (cause by confusing presentation) > cognitive capacity

- Problem: **One or both channels are overloaded** by essential and incidental processing demands (cause by confusing presentation) .
- Solution 6: **Aligning** words & pictures**:** Place printed words near corresponding parts of graphics to reduce need for visual scanning.
- Solution 7: **Eliminating Redundancy:** Avoid presenting identical streams of printed and spoken words.

Example of Separated Presentations of words and pictures

Example of Aligning words and pictures

Type 4 - Essential processing + incidental processing (cause by confusing presentation) > cognitive capacity

- Test the effectiveness of **Aligning**:
- Moreno and Mayer (1999) found that students who learned from integrated presentations (consisting of animation with integrated on-screen text) performed better on a problem-solving transfer test than did students who learned from separated presentations (consisting of animation with separated on-screen text).（使用整合動畫與螢幕文字的畫面，學生的問題解決分數較佳）
- The effect size was 0.48.
- This result is referred to as a **Spatial Contiguity Effect**.

Type 4 - Essential processing + incidental processing (cause by confusing presentation) > cognitive capacity

- Test the effectiveness of **Eliminating Redundancy**:
- Situation of redundant presentation:

  animation + narration + on-screen text.
- In a series of three studies, students who learned from nonredundant presentations performed better on problem-solving transfer tests than did students who learned from redundant presentations. （學生使用無冗餘的教材，其問題解決分數較佳）
- The median effect size was 0.69.
- This result is referred to as a **Redundancy Effect**.

Type 5 - Essential processing + representational holding > cognitive capacity (the system is overloaded by the need to hold information in working memory)

- Problem: **One or both channels are overloaded** by essential processing and representational holding.
- Solution 8: **Synchronizing**: Present narration and corresponding animation simultaneously to minimize need to hold representations in memory.
- Solution 9: **Individualizing**: Make sure learners possess skill at holding mental representations.

Type 5 - Essential processing + representational holding > cognitive capacity

- Test the effectiveness of **Synchronizing :**
- In a series of eight studies (Mayer & Anderson, 1991; etc.), students performed better on tests of problem-solving transfer when they learned from simultaneous presentations than from successive presentations.（學生使用同時性的教材呈現，其問題解決表現高於使用連續性教材呈現的學生）
- The median effect size was 1.30.
- We refer to this result as a **Temporal Contiguity Effect**.

Type 5 - Essential processing + representational holding > cognitive capacity

- Test the effectiveness of **Individualizing**:
- Mayer and Sims (1994) found that high-spatial learners performed much better on problem-solving transfer tests from simultaneous presentation than from successive presentation, whereas low-spatial learners performed at the same low level for both. （高空間能力學習者在同時性的教材呈現比連續性教材呈現的表現較佳，低空間能力者在兩者的表現皆較差）
- The median effect size was 1.13.
- The researchers refer to this interaction as a **Spatial Ability Effect**.

The Challenge of Designing Instruction to Reduces Cognitive Load

- **A major challenge**: meaningful learning can require a heavy amount of essential cognitive processing, but the cognitive resources of the learner's information processing system are severely limited.
- Multimedia instruction should be designed in ways that **minimize any unnecessary cognitive load**.

The Challenge of Designing Instruction to Reduces Cognitive Load

- Effective instructional design depends on **sensitivity to cognitive load** which, in turn, depends on an understanding of **how the human mind works**.
- In this article, the authors shared the fruits of 12 years of programmatic research at UCSB (Univ. of California, Santa Barbara), aimed at contribution to cognitive theory and building an empirical database.